

Solvable $\mathcal{PT}$ -Symmetric Hamiltonians*

M. Znojil**

Nuclear Physics Institute AV ČR, CZ-250 68 Řež, Czech Republic

Received June 6, 2001

Abstract—Within the so-called $\mathcal{PT}$ -symmetric version of quantum mechanics, a brief review of exactly solvable models is given. The distinction is made between curved and straight coordinate lines, between their unbounded (aperiodic) and bounded (periodic) choices, and between completely and partially solvable cases. © 2002 MAIK “Nauka/Interperiodica”.

1. INTRODUCTION

$\mathcal{PT}$ -symmetric quantum mechanics [1] was independently proposed and used as a methodological laboratory in quantum physics by several groups of authors: by Caliceti *et al.* [2] in perturbation theory, by Bessis *et al.* [3] in field theory, and by Andrianov *et al.* [4] in the supersymmetric context.

This short review will pay attention to (partially or completely) exactly solvable models within this framework, with emphasis on the results obtained by the present author.

2. COMPLETE SOLVABILITY ON CURVED PATHS

One of the first exactly and completely solvable examples of a $\mathcal{PT}$ -symmetric system was found by Cannata *et al.* [4] and rediscovered by Bender *et al.* [5] more than one year later. Its modified Schrödinger bound-state problem is defined on certain curved, left–right-symmetric “generalized coordinate” lines in a complex plane. Mathematically, it is defined via an exponential potential and proves to be exactly solvable in terms of Bessel functions. Owing to its relationship to a power-law force in the large-exponent limit, it can be most simply interpreted as a certain smooth and non-Hermitian $\mathcal{PT}$ -symmetric analog of the current square well.

Recently, the double-well counterpart of the latter set of models was shown to be exactly solvable, in terms of Laguerre polynomials, in [6]. In a way similar to the above “single-well” example, its paths of integration are the same, curved complex lines again. Their spectra exhibit a puzzling and highly

unexpected feature of certain coupling-dependent rearrangements mediated by “unavoided” crossings at critical points. This phenomenon reflects the non-Hermiticity of the Hamiltonian [7].

Via a suitable Liouvillean change of variables in the above double-well-like differential Schrödinger equation, one can immediately obtain another Laguerre-related solvable system with a potential of the Coulombic single-pole form. In this case, the deformation of the integration path plays the beneficial role of a natural regularization prescription. At the same time, it also leads to the need for working with complex charges in a way described in [8]. Also, the related energy spectrum exhibits certain unexpected features: positivity, the coexistence of the growth and decrease with increasing coupling strength, etc.

There is presently no clear mathematical explanation of this behavior; a consistent and/or possible physical interpretation of counterintuitive models of this type has not been developed either. A much better situation emerges in the case of integration curves defined as left–right-symmetric straight lines.

3. STRAIGHT PATHS

By using the language of the so-called Kustaanheimo–Stiefel transformation (see [9] for detailed references), the above two curvilinear examples can be shown to be equivalent to the $\mathcal{PT}$ -symmetric harmonic oscillator of [10] with a centrifugal term regularized by a mere downward complex shift of the (full) real axis,

$$\left(-\frac{d^2}{dx^2} + x^2 - 2icx + \frac{\alpha^2 - 1/4}{(x - ic)^2}\right)\varphi(x) = (E + c^2)\varphi(x), \quad \varphi(x) \in L_2(-\infty, \infty). \quad (1)$$

This oscillator with the Laguerre polynomial normalizable solutions

$$\varphi_{(\pm n)}(x) = \mathcal{N}(x - ic)^{\pm\alpha+1/2} e^{-(x-ic)^2/2}$$

*This article was submitted by the author in English.

**e-mail: znojil@ujf.cas.cz

$$\times L_n^{(\pm\alpha)}[(x-ic)^2], \quad n=0,1,\dots,$$

possesses the nonequidistant energy spectrum $E = E_{(\pm n)} = 4n + 2 \pm 2\alpha$ and represents a certain unperturbed limit of the quartic oscillator models formulated by Buslaev and Grecchi [11]. Unfortunately these authors did not notice the existence of the “quasieven,” $(-n)$ -signed half of the spectrum. This omission can be easily corrected. One just introduces a “two-to-one” isospectrality correspondence between the respective Hermitian and non-Hermitian anharmonic oscillator models of [11].

Interpretation of models living on straight lines becomes significantly facilitated by the easier identification and interpretation of their complex components [12]. Immediate purely analytic constructions recover, e.g., the existence of models that are in a one-to-one correspondence with the so-called shape-invariant real forces in one dimension (compare with their presentation in [13]) and on the half-line (their $\mathcal{PT}$ -symmetric counterparts were described and listed in [14]).

The situation is reviewed in [15]. A fully general form of this type of analytic constructions dates back to the introduction of the so-called Natanzon potentials and, in the present context, is thoroughly analyzed and described in [16].

4. MODELS WITH PERIODIC BOUNDARY CONDITIONS

A new and promising development of $\mathcal{PT}$ -symmetric considerations has been recently inspired by the study of two- and three-particle models [17]. The $\mathcal{PT}$ symmetrization of the Hamiltonians has been again conjectured to be sufficient for keeping their spectrum real. The related “weakening of the Hermiticity” finds a natural generalization in the new context.

Particular attention was paid to the possible non-Hermitian generalizations of the well-known Calogero model [18]. In this setting, the separability of the underlying partial differential Schrödinger equation in hyperspherical coordinates helps us to reduce the problem to a “hyperangular” ordinary differential equation defined on a finite interval. In this way, one has, in the simplest cases, to solve the complexified ordinary differential equations of the generalized Pöschl–Teller type,

$$\left(-\frac{d^2}{d\phi^2} + \frac{l(l+1)}{\sin^2 \phi} + \frac{\lambda(\lambda+1)}{\cos^2 \phi}\right) \chi(\phi) = E\chi(\phi), \quad (2)$$

on an interval $\phi \in (-M\pi/2, M\pi/2)$ with a suitable integer M . These equations can be solved exactly in terms of hypergeometric functions [19].

Strongly repulsive singularities at $\phi_j = j\pi/2$ are currently not penetrable [20]. Here, they become regularized in a $\mathcal{PT}$ -symmetric manner, which parallels a few older constructions on unbounded intervals [14]. Quasisymmetric and quasiantisymmetric solutions arise from certain *ad hoc* boundary conditions [17, 21].

For the most elementary illustration, let us now set $\lambda = 0$ and $M = 2$. Then, the differential Eq. (2) possesses two independent hypergeometric solutions,

$$\chi^{(\pm)}(\phi) = (\sin \phi)^{1/2 \pm \alpha} {}_2F_1(u^{(\pm)}, v^{(\pm)}; 1 \pm \alpha; \sin^2 \phi),$$

$$\alpha = l + 1/2 > 0,$$

where $2u^{(\pm)} = 1/2 - \beta \pm \alpha$ and $2v^{(\pm)} = 1/2 + \beta \pm \alpha$. At the boundary of convergence $\sin^2 \phi = 1$, the matching of the logarithmic derivatives is equivalent to the termination of this series, whereupon there arise Gegenbauer polynomials,

$$\chi(\phi) = \chi_{(\pm k)}(\phi) = (\sin \phi)^{1/2 \pm \alpha} C_k^{1/2 \pm \alpha}(\cos \phi),$$

$$k = 0, 1, \dots$$

The construction also quantizes the energies and gives them in the closed form

$$E = E_{(\pm k)} = (k \pm \alpha + 1/2)^2, \quad k = 0, 1, \dots$$

This set of eigenvalues is composed of the two subsets in a way that resembles the above-mentioned nonequidistant spectrum of the $\mathcal{PT}$ -symmetrized singular harmonic oscillator (1) [10].

5. CONCLUSIONS AND OUTLOOK

Historically, one of the first persuasive manifestations of the merits and power of $\mathcal{PT}$ symmetry was offered by Bender and Boettcher [22], who discovered the quasiexact (i.e., incomplete) solvability of the most common and popular quartic polynomial oscillators.

The plausible reasons for the unexpected delay of such an “obvious” observation are closely related to the above-mentioned “forgotten” energies. One has to keep in mind the “spontaneous” regularity of singular potentials within the new formalism. This observation was made explicit in [23], where the presence of two additional singular terms was shown to be compatible with the quasiexact solvability of quartic potentials.

In the quasiexact context, the changes of variables can play the same role as in the completely solvable models. This was illustrated by the particular constructions of the decadic model [24] and of the harmonic plus Coulomb superposition [25]. Further work in this direction is in progress [26].

ACKNOWLEDGMENTS

This work was supported by the Grant Agency of the Academy of Sciences of the Czech Republic (contract no. A 1048004).

REFERENCES

1. C. M. Bender, S. Boettcher, K. A. Milton, and P. N. Meisinger, *J. Math. Phys. (N.Y.)* **40**, 2201 (1999).
2. E. Calicetti, S. Graffi, and M. Maioli, *Commun. Math. Phys.* **75**, 51 (1980); G. Álvarez, *J. Phys. A* **27**, 4589 (1995); M. F. Fernández, R. Guardiola, J. Ros, and M. Znojil, *J. Phys. A* **31**, 10105 (1998).
3. D. Bessis and J. Zinn-Justin, private communication (1992); C. M. Bender and A. Turbiner, *Phys. Lett. A* **173**, 442 (1993); C. M. Bender and K. A. Milton, *Phys. Rev. D* **55**, R3255 (1997); C. M. Bender and S. Boettcher, *Phys. Rev. Lett.* **80**, 5243 (1998); E. Delabaere and F. Pham, *Phys. Lett. A* **250**, 25, 29 (1998).
4. A. A. Andrianov, F. Cannata, J.-P. Dedonder, and M. V. Ioffe, *Int. J. Mod. Phys. A* **14**, 2675 (1999); F. Cannata, G. Junker, and J. Trost, *Phys. Lett. A* **246**, 219 (1998); B. Bagchi and R. Roychoudhury, *J. Phys. A* **33**, L1 (2000); M. Znojil, F. Cannata, B. Bagchi, and R. Roychoudhury, *Phys. Lett. B* **483**, 284 (2000).
5. C. M. Bender, S. Boettcher, H. F. Jones, and Van M. Savage, *J. Phys. A* **32**, 6771 (1999).
6. M. Znojil, *Phys. Lett. A* **264**, 108 (1999).
7. I. Herbst, private communication.
8. M. Znojil and G. Levai, *Phys. Lett. A* **271**, 327 (2000).
9. P. Kustaanheimo and E. Stiefel, *J. Reine Angew. Math.* **218**, 204 (1965); E. G. Kalnins, W. Miller, Jr., and G. S. Pogosyan, *J. Math. Phys. (N.Y.)* **41**, 2629 (2000).
10. M. Znojil, *Phys. Lett. A* **259**, 220 (1999).
11. V. Buslaev and V. Grecchi, *J. Phys. A* **26**, 5541 (1993).
12. M. Znojil, *J. Phys. A* **32**, 7419 (1999); M. F. Fernández, R. Guardiola, J. Ros, and M. Znojil, *J. Phys. A* **32**, 3105 (1999).
13. M. Znojil, *J. Phys. A* **33**, L61 (2000).
14. M. Znojil, *J. Phys. A* **33**, 4561 (2000).
15. M. Znojil, *Rend. Circ. Mat. Palermo Serie II, Suppl.* **66**, 213 (2001).
16. G. Levai and M. Znojil, *J. Phys. A* **33**, 7165 (2000).
17. M. Znojil and M. Tater, *J. Phys. A* **34**, 1793 (2001).
18. F. Calogero, *J. Math. Phys. (N.Y.)* **10**, 2191 (1969); A. Turbiner, *Mod. Phys. Lett. A* **13**, 1473 (1998) and references therein.
19. F. Cooper, A. Khare, and U. Sukhatme, *Phys. Rep.* **251**, 267 (1995).
20. S. Flügge, *Practical Quantum Mechanics I* (Springer-Verlag, Berlin, 1971), p. 89.
21. M. Znojil, *Phys. Rev. A* **61**, 066101 (2000).
22. C. M. Bender and S. Boettcher, *J. Phys. A* **31**, L273 (1998).
23. M. Znojil, *J. Phys. A* **33**, 4203 (2000).
24. M. Znojil, *J. Phys. A* **33**, 6825 (2000).
25. M. Znojil, *J. Phys. A* **32**, 4563 (1999).
26. F. Gemperle, M. Znojil, and F. Cannata, in preparation.